Solving Equations

Practice Worksheet — One-Step, Two-Step, Multi-Step & Word Problems

Level: Pre-Algebra / Algebra 1

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Instructions: Show all of your work for full credit. Study the worked example at the start of each section before you begin that section's problems.

PART 1 — ONE-STEP EQUATIONS (10 problems)

Worked Example — Follow These Steps

Solve for x: $x + 7 = 12$

$x + 7 = 12$	Given equation
$x + 7 - 7 = 12 - 7$	Subtract 7 from each side (undo the addition)
$x = 5$	Simplify

Remember: addition and subtraction are inverse operations; multiplication and division are inverse operations. Undo whatever is being done to x.

Solve each equation for x. Show your work in the space provided.

1. $x + 9 = 14$

2. $x - 6 = 8$

3. $x - 4 = -9$

4. $7 + x = 2$

5. $5x = 35$

6. $-3x = 21$

7. $\frac{x}{4} = 9$

8. $\frac{x}{6} = -3$

9. $-x = 9$

10. $12 = x + 5$

PART 2 — TWO-STEP EQUATIONS (10 problems)

Worked Example — Follow These Steps

Solve for x: $3x + 4 = 19$

$3x + 4 = 19$	Given equation
$3x + 4 - 4 = 19 - 4$	Subtract 4 from each side (undo the addition)
$3x = 15$	Simplify
$\frac{3x}{3} = \frac{15}{3}$	Divide each side by 3 (undo the coefficient)
$x = 5$	Simplify

Remember: undo addition/subtraction *first*, then undo multiplication/division.

Solve each equation for x. Show your work in the space provided.

1. $2x + 5 = 17$

2. $3x - 7 = 11$

3. $4x + 9 = -7$

4. $-2x + 6 = 14$

5. $5x - 3 = -18$

6. $-6x - 4 = 20$

7. $\frac{x}{2} + 3 = 10$

8. $\frac{x}{3} - 5 = -2$

9. $-\frac{x}{4} + 6 = 2$

10. $7 - 2x = -5$

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PART 3 — MULTI-STEP EQUATIONS WITH FRACTIONS & DECIMALS (10 problems)

Worked Example — Follow These Steps

Solve for x: $2(x-3)+1.5=9.5$

$2(x-3)+1.5=9.5$	Given equation
$2x-6+1.5=9.5$	Distribute the 2
$2x-4.5=9.5$	Combine like terms
$2x-4.5+4.5=9.5+4.5$	Add 4.5 to each side
$2x=14$	Simplify
$x=7$	Divide each side by 2

Tips

Fractions: multiply both sides by the denominator (or the LCD, if there is more than one fraction) to clear it before doing anything else. **Decimals:** treat them just like whole numbers — add, subtract, multiply, and divide the same way, just watch your decimal point.

Solve each equation for x. Show your work in the space provided.

1. $0.5x+3=11$

2. $\frac{x}{3}-2=5$

3. $2x+0.75=4.75$

4. $\frac{x+5}{4}=3$

5. $0.2x-1=0.6x-9$

6. $\frac{2}{3}x+1=7$

7. $3(x-1.5)=10.5$

8. $\frac{x}{5}+\frac{x}{2}=7$

9. $0.25(x+8)=5$

10. $\frac{3x-2}{5}=2$

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PART 4 — WORD PROBLEMS (5 problems)

Worked Example — Follow These Steps

The sum of a number and 8 is 23. Find the number.

Let $x =$ the unknown number.	
$x + 8 = 23$	Translate the sentence into an equation
$x + 8 - 8 = 23 - 8$	Subtract 8 from each side
$x = 15$	Simplify

The number is 15.

Tip: underline the key phrase, define what x represents in words, write the equation, solve it, then answer the original question in a full sentence.

Solve each word problem below. Define your variable, write an equation, and show all your work.

1. Five more than three times a number is 26. Find the number.

2. The sum of two consecutive integers is 45. Find the two integers.

3. Maria is 4 years older than twice her brother's age. If Maria is 22 years old, how old is her brother?

4. A gym membership charges a \$20 sign-up fee plus \$15 per month. If Sam paid a total of \$110, how many months did he pay for?

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5. The length of a rectangle is 3 cm more than twice its width. If the perimeter of the rectangle is 36 cm, find the width and the length.

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Answer Key

Part 1 — One-Step Equations

1. $x = 5$ 2. $x = 14$ 3. $x = -5$ 4. $x = -5$ 5. $x = 7$ 6. $x = -7$ 7. $x = 36$ 8. $x = -18$ 9. $x = -9$ 10. $x = 7$

Part 2 — Two-Step Equations

1. $x = 6$ 2. $x = 6$ 3. $x = -4$ 4. $x = -4$ 5. $x = -3$ 6. $x = -4$ 7. $x = 14$ 8. $x = 9$ 9. $x = 16$ 10. $x = 6$

Part 3 — Multi-Step Equations with Fractions & Decimals

1. $x = 16$ 2. $x = 21$ 3. $x = 2$ 4. $x = 7$ 5. $x = 20$ 6. $x = 9$ 7. $x = 5$ 8. $x = 10$ 9. $x = 12$ 10. $x = 4$

Part 4 — Word Problems

1. Let x = the number. $3x + 5 = 26 \Rightarrow 3x = 21 \Rightarrow x = 7$. **The number is 7.**

2. Let x = the first integer, $x + 1$ = the second. $x + (x + 1) = 45 \Rightarrow 2x + 1 = 45 \Rightarrow 2x = 44 \Rightarrow x = 22$. **The integers are 22 and 23.**

3. Let x = her brother's age. $2x + 4 = 22 \Rightarrow 2x = 18 \Rightarrow x = 9$. **Her brother is 9 years old.**

4. Let x = the number of months. $15x + 20 = 110 \Rightarrow 15x = 90 \Rightarrow x = 6$. **Sam paid for 6 months.**

5. Let w = the width, so the length is $2w + 3$. Perimeter: $2(w + (2w + 3)) = 36 \Rightarrow 2(3w + 3) = 36 \Rightarrow 6w + 6 = 36 \Rightarrow 6w = 30 \Rightarrow w = 5$. Length $= 2(5) + 3 = 13$. **Width** $= 5$ **cm, Length** $= 13$ **cm.**

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